

MoodCast - App Summary (Repo-Based)

What it is

MoodCast is an iOS app that curates podcast episodes from Spotify based on user mood. It also captures route/ETE context and supports playback and favorites management.

Who it is for

Primary persona: commuters/drivers who want mood-aligned podcast picks for a drive.

What it does

- Creates accounts and signs users in with Firebase Authentication.
- Stores user profile data and app state in Firestore.
- Links a Spotify account via OAuth callback flow.
- Generates an episode stack from a mood via backend Spotify search endpoints.
- Supports swipe interactions to favorite or dismiss episode cards.
- Lets users search saved/favorited episodes by name/description.
- Provides playback controls (play, pause, seek, status) through backend API routes.

How it works (architecture)

Components/services

- iOS SwiftUI client: views + view models (opening, deck, search, profile, account).
- Client services: AuthService, DataService, SpotifyService.
- Backend: Node.js Express server (``glitch_server/server.js``).
- Data/auth: Firebase Auth + Firestore (``firebase-admin``, iOS Firebase SDK).
- External content: Spotify Web API (``spotify-web-api-node``) and Spotify OAuth.
- Optional ranking/filtering path: OpenAI call exists in server code, but is commented out.

Data flow

- User actions in SwiftUI call DataService/AuthService methods.
- iOS app sends HTTP requests to ``https://cscd-488-project.glitch.me``.
- Backend reads/writes user docs and tokens in Firestore.
- Backend calls Spotify APIs (refresh token when needed), returns JSON to iOS app.

How to run (minimal getting started)

1. Backend setup

- Go to ``MoodCast_Final_Deliverables/glitch_server``.
- Install dependencies: ``npm install``.
- Add ``.env`` values used in server code: ``SPOTIFY_CLIENT_ID``, ``SPOTIFY_CLIENT_SECRET``, ``SPOTIFY_REDIRECT_URI``, ``OPENAI_API_KEY``.
- Add Firebase service account JSON as ``serviceAccountKey.json``.
- Start server: ``npm start`` (defaults to port 3000 unless ``PORT`` is set).

2. iOS app setup

- Open ``MoodCast_Final_Deliverables/MoodCast/MoodCast.xcodeproj`` in Xcode.
- Ensure Firebase is configured (``GoogleService-Info.plist`` present) and app

URL type

- matches Spotify redirect (example in README: ``myapp://spotify/callback``).
- Build and run the app from Xcode.

Not found in repo

- A definitive production deployment recipe for the backend host.
- A command-line iOS build/run script (Xcode GUI flow is implied).